FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 231**

General Instructions:

- The test consists of total 69 questions.
- Each subject (PCM) has 23 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-C & Section-D.**

Section-A (01 – 03, 24 – 26, 47 – 49) *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer.*
Partial Marks **+1** for each correct option provided no incorrect options is selected.

Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Section-C (11 – 20, 34 – 43, 57 – 66) *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Section-D (21 – 23, 44 – 46, 67 – 69) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

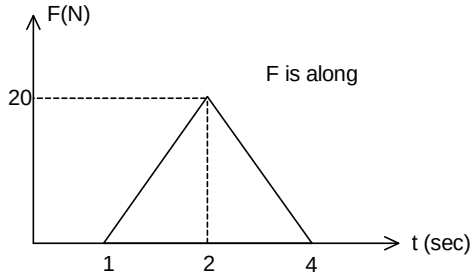
Physics

PART – I

SECTION – A
(Single Correct Choice Type)

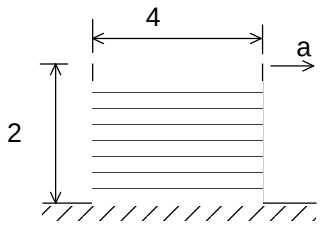
This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. A particle of mass 2 kg is moving along x-axis with a velocity of 5 m/s. It crosses the origin at t = 0. A time varying force whose variation is as shown starts acting on a particle at t = 1 sec. For this situation, mark out the correct statement.



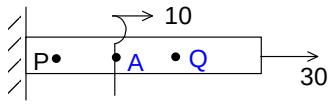
(A) the particle will come to rest instantaneously at t = 6.8 sec
(B) the final velocity of the particle will be 18 m/s.
(C) work done by the force F on the particle is 375 J.
(D) all of the above

2. A liquid is filled in a rectangular vessel of dimension 4 m $\times$ 3 m $\times$ 2 m. Now the container starts to move with uniform acceleration 1.25 m/s^2 at t = 0. The volume of liquid in vessel is $18m^3$. The speed of liquid coming out from a very small orifice made at bottom of right side wall at t = 4s is



(A) zero
(B) $\sqrt{30}$ m / s
(C) 5 m/s
(D) 10 m/s

3. One end of a rope is bound to a vertical wall. The other end is pulled with a horizontal force of 30 N. An additional force of 10 N is applied to the right at the point A in the middle of the rope. Which of the following is true about tensions at points P and Q



(A) $T_P = T_Q = 40N$
(B) $T_P = T_Q = 20N$
(C) $T_P - T_Q = 10N$
(D) $T_Q = 20N, T_P = 40N$

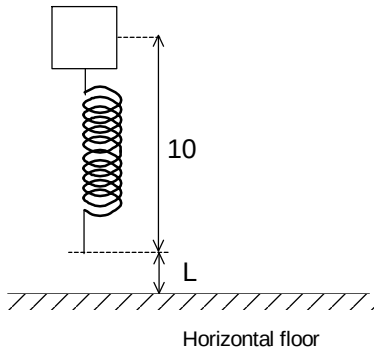
Space for Rough work

(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

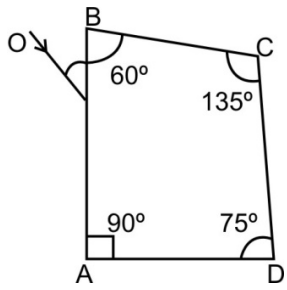
4. With regard to sound waves, mark out the correct statement(s).
   (A) the phase difference between incident & reflected displacement waves from the fixed end is zero.
   (B) the phase difference between incident & reflected pressure waves from the fixed end is zero.
   (C) the phase difference between incident & reflected pressure waves from the fixed end is π.
   (D) the phase difference between incident & reflected pressure waves from the free end is π.

5. A small block mass m is fixed at upper end of a massless vertical spring of spring constant K = $\frac{4mg}{L}$ and natural length 10 L. The lower end of spring is free and is at a height L from fixed horizontal floor as shown. The spring is initially unstretched and the spring block system is released from rest in the shown position.



   (A) When the speed of block is maximum the magnitude of force exerted by spring on the block is mg.
   (B) The maximum speed attained by the block is $\frac{3}{2}\sqrt{gL}$.
   (C) The maximum compression of the spring is L/8.
   (D) The time taken by the mass to move from the equilibrium position to the maximum compressed position of the spring is $\frac{\pi}{4}\sqrt{\frac{L}{g}}$

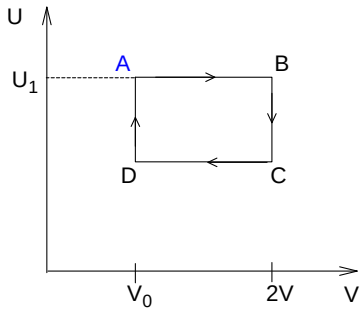
6. A ray OP of monochromatic light is incident on the face AB of prism ABCD near vertex B at an incident angle of 60º as shown in figure. If the refractive index of the material of the prism is $\sqrt{3}$, which of the following is(are) correct ?



   (A) The ray gets totally internally reflected at face CD.
   (B) The ray comes out through face AD.
   (C) The angle between the incident ray and the emergent ray is 90º
   (D) The angle between the incident ray and the emergent ray is 120º

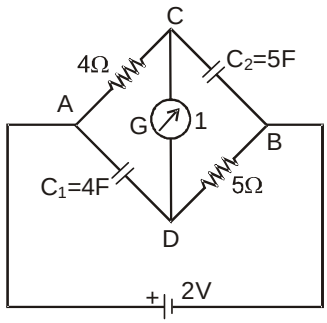
Space for Rough work

7. One mole of an ideal gas has an internal energy given by U = U_o + 2 PV, where P is the pressure and V the volume of the gas, U_o is a constant. this gas undergoes the quasi-static cyclic process ABCD as shown the U-V diagram. choose the correct statement (s) :



(A) The work done by the ideal gas in the process AB is $\frac{U_1 - U_0}{2} \ln 2$.
(B) The gas must be monoatomic
(C) The gas must be diatomic.
(D) The molar heat capacity of the gas at constant pressure is 3 R

8. In the circuit shown in Fig. the cell is ideal, with e.m.f. = 2V. The resistance of the coil of the galvanometer G is 1Ω. Then



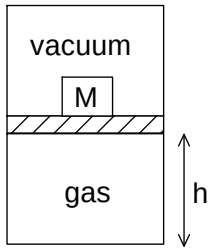
(A) No current flows in G
(B) 0.2 A current flows in G
(C) Potential difference across C_1 is 1V
(D) Potential difference across C_2 is 1.2 V

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 9 & 10

One mole of an ideal diatomic gas is contained in the lower part of the cylinder beneath a piston of area 50 cm^2 as shown in the figure. The piston has negligible mass but a block of mass M = 100 kg is kept on the piston. Initially the gas is maintained at a temperature of 300 K and piston is at a height h from the bottom of the cylinder. The gas is now heated to raise the weight by 10 cm. [Take g = 10 m/s^2, atmospheric pressure = 10^5 N/m^2]



9. The thermal energy supplied to the gas in the process is
(A) 249 J (B) 348.6 J
(C) 99.6 J (D) 100 J

Space for Rough work

10. If a small hole has been created in the upper part of the cylinder, then the value of h before heating the gas is
(A) 1.66 m (B) 2.49 m
(C) 1.40 m (D) none of these

SECTION – C
(Single digit integer type)

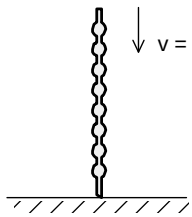
This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

11. A container filled with air under pressure P_0 contains a soap bubble of radius R. When the air pressure is reduced to half isothermally, the bubble radius becomes (5R/4). If the surface tension of the soap water solution is S, then find $\left(\frac{RP_o}{12S}\right)$.

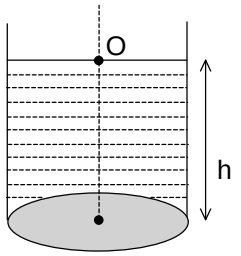
12. A glass tube of 1.0 m length is filled with water. The water is drained out very slowly from the tube through a hole in the bottom while a vibrating 500 HZ tuning fork is held near the open upper end of the tube. If the speed of sound is 320 ms^{-1}, find the number of resonance that can be obtained.

13. A 50 Ω resistor is connected in series with a coil and a sinusoidal ac–source of effective voltage 50 V. If the effective voltage drop across the resistor is 20 V and across the coil is 40 V, Find the power developed in coil in watt?

14. The upper end of the uniform chain of linear mass density 2 kg/m is lowered with a speed 1 m/s as shown in diagram. The normal reaction offered by the ground after the chain fall through a distance 5 cm is given by k newton. Find the value of k.

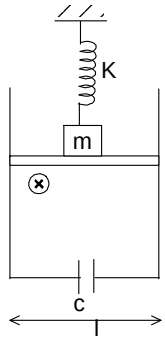


15. Bottom of glass beaker is made of a thin equi-convex lens having bottom side silver polished as shown in the figure. Now the water is filled in the beaker upto a height of h = 4 m. The image of point object floating at middle point of beaker at the surface of water coincides with it. The value of radius of curvature of lens is $\left(\mu_g = \frac{3}{2}, \mu_w = \frac{4}{3}\right)$

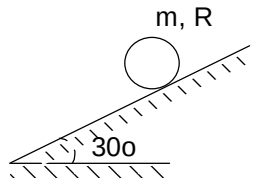


Space for Rough work

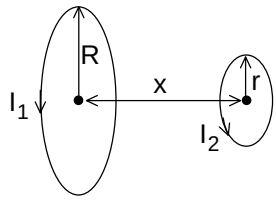
16. A block is attached to the ceiling by a spring that has a force constant K = 200 N/m. A conducting rod is rigidly attached to the block. The combined mass of the block and the rod is m = 0.3 kg. The rod can slide without friction along the two vertical parallel rails, which are a distance ℓ = 1 m apart. A capacitor of known capacitance C = 500 μF is attached to the rails by the wires. The entire system is placed in a uniform magnetic field B = 20 T directed inwards as shown. Neglect the self inductance and electrical resistance of the rod and all wires. The angular frequency (in rad/s) of the vertical oscillations of the block is 4ω. Find the value of ω?



17. A uniform spherical shell of mass m and radius R starts from rest with pure rolling on a long inclined plane as shown in the figure. If angular momentum of shell about point of contact after 1 sec of its starting is KmR, find K [Take g = 10 m/s^2] ……

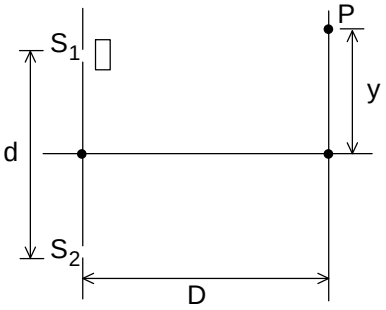


18. The force of interaction between two co-axial current loops having radii R & r (r<<R), separated by a distance x carrying currents I_1 & I_2 respectively is given by the expression, $F = \dfrac{N\mu_o \pi R^2 r^2 I_1 I_2 x}{2\left(R^2 + x^2\right)^{\frac{5}{2}}}$. Find 'N'



19. The minimum deviation suffered by a ray emerging through a prism $\left(\mu = \dfrac{2}{\sqrt{3}}\right)$ is equal to half the refracting angle of the prism. Find the value of C if the refracting angle of the prism is $(2\pi / C)$.

20. Young's double slit experiment is conducted with white light (4000A^o –7000A^o). As shown in the diagram, the slit separation $d = \left(\dfrac{1}{2}\right)$mm, and the screen slit distance, D = 1 m. The upper slit is covered with a film of thickness, t = 1 μm having a refractive index, $\mu = \dfrac{3}{2}$. Find the number of wavelengths which will form a maxima at y = 3mm

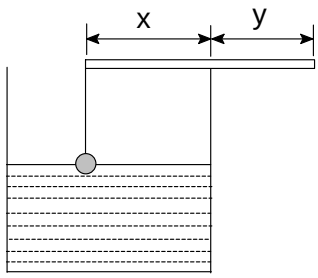


Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

21. A homogeneous aluminium ball of radius r = 0.5 cm is suspended on a weightless thread from an end of a homogeneous rod of mass M = 4.4 g. The rod is placed on the edge of a conditioner with water so that half of the ball is submerged in water when the system is in equilibrium as shown. The densities ρ_{al} and ρ_w of aluminium an water are 2.7×10^3 kg/m^3 and 1×10^3 kg/m^3 respectively. Determine the ratio y/x of the parts of the rod to the brim, neglecting the surface tension on the boundaries between the ball and water.



22. A parachutist jumps from the top of a very high tower with a siren of frequency 800 Hz on his back. Assume his initial velocity to be zero. After falling freely for 12s, he observes that the frequency of sound heard by him reflected from level ground below him is differing by 700 Hz w.r.t. the original frequency. What was the height of tower. Velocity of sound in air is 330 m/s, and g = 10 m/s^2.

23. A radionuclide with disintegration constant λ is produced in a reactor at a constant rate $\alpha(=2\lambda)$ nuclei per second. At t = 0, there are no nuclei present in the reactor. During each decay energy E_0 is released. 20% of this energy is utilized in increasing the temperature of water. Find the increase in temperature of m mass of water in time t = $T_{1/2}$ if specific heat of water is s. Assume that there is no loss of energy through water surface. (Given E_0 = 100 times the energy required to raise the temperature of mass m of water from 0^0 to 100^0C).

Space for Rough work

Chemistry

PART – II

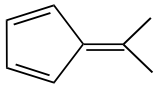
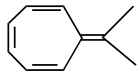
SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

24. Number of S–S bond in $H_2S_nO_6$

(A) n (B) (n–1)
(C) (n–2) (D) (n + 1)

25. Which one of the following statements is True:

(1)  (2) 

(A) PhLi adds to both compound with equal ease
(B) PhLi does not add to either of the compound
(C) PhLi react readily with 1 but does not add to 2.
(D) PhLi react readily with 2 but does not add to 1.

26. One mole of Argon is heated using $PV^{5/2}$ = constant. By what amount heat is absorbed during the process, when temperature changes by $\Delta T = 26$ K.

(A) 100 J (B) 200 J
(C) 180 J (D) 208 J

(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

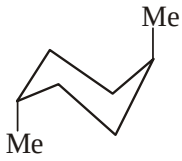
27. Select the correct statement(s).
(A) For equilibrium, pressure is constant.
(B) For equilibrium, temperature is constant.
(C) For equilibrium, concentration is constant.
(D) For equilibrium, pressure & temperature is variable.

Space for Rough work

28. A_2O_n is oxidised to AO_3^- by MnO_4^- in acid medium. If 1.5×10^{-3} mole of A_2O_n requires 60 ml of 0.03 M-$KMnO_4$ solution for complete oxidation, which of the following statement(s) is/are correct?
(A) The value of n = 2
(B) Empirical formula of oxide is AO.
(C) 0.15 mole of A_2O_n would require 0.15 mole of acidified $K_2Cr_2O_7$ solution
(D) 'A" may be a metal from second group of periodic table.

29. Select the correct statement for the following species.

(I) 

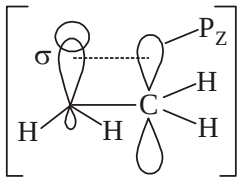
(II) 

(III) 

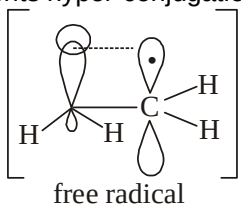
(IV) 

(A) I and III are Geometrical isomers.
(B) I and II are Conformational pairs
(C) III and IV are Conformational pairs
(D) II and IV are Geometrical isomers

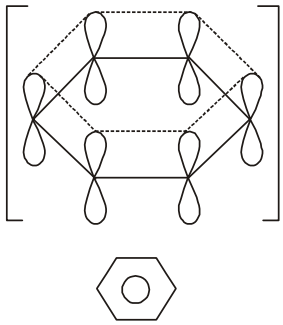
30. Among the following orbital interactions which represents hyper-conjugation interaction?

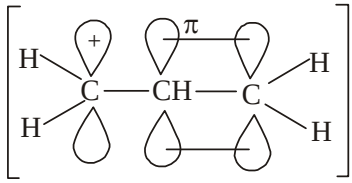
(A) 

$CH_3–CH_2^{\oplus}$

(B) 

free radical
$(CH_3–\dot{C}H_2)$

(C) 

(D) 

$(H_2\overset{\oplus}{C} – CH = CH_2)$

Space for Rough work

31. Which of the following statements is/are false
(A) In $[PtCl_2(NH_3)_4]^{2+}$ complex ion, the cis-form is optically active, while trans-form is optically inactive
(B) In $[Fe(C_2O_4)_3]^{3-}$, geometrical isomerism does not exist, while optical isomerism exists
(C) $[Mabcd]^{n\pm}$ square planar complexes exhibit both optical as well as geometrical isomerism
(D) In $[Mabcd]^{n\pm}$ tetrahedral complexes, optical isomerism cannot be observed

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 32 & 33

Complex compounds which have same molecular formula but have different structural arrangements of ligands around central metal atom or ion are called structural isomers and phenomenon is named as structural isomerism.

32. Which of the following compounds is/are polymerisation isomer of $[Fe(NO_2)_3(NH_3)_3]$.
(I) $[Fe(NO^7)(NH_3)_5]$ $[Fe(NO_2)_5(NH_3)]$
(II) $[Fe(NO_2)_2(NH_3)_4]_2$ $[Fe(NO_2)_5(NH_3)]$
(III) $[Fe(NO_2)(NH_3)_5]$ $[Fe(NO_2)_4(NH_3)_2]_2$
(IV) $[Fe(NO_2)_2(NH_3)_4]$ $[Fe(NO_2)_4(NH_3)_2]$
Choose the correct code :
(A) II, III
(B) II, III, IV
(C) I, IV
(D) I, II, III, IV

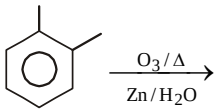
33. Select incorrect match
(A) $[Co(NO_2)(H_2O)(en)_2)]Cl_2$, $[CoCl(NO_2)(en)_2]$ Cl . H_2O – Hydrate isomerism
(B) $[Cu(NH_3)_4][PtCl_4]$, $[CuCl_2(NH_3)_2]$ $[PtCl_2(NH_3)_2]$ – Coordination isomerism
(C) $[Ni(CN)(H_2O)(NH_3)_4]Cl$, $[NiCl(H_2O)(NH_3)_4]CN$ – Ionization isomerism
(D) $[Cr(NCS)(NH_3)_5]$ $[ZnCl_4]$, $[Cr(SCN)(NH_3)_5]$ $[ZnCl_4]$ – Linkage isomerism

Space for Rough work

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

34. How many more coordination isomers are possible of the following complex compound.
$[PtCl_2 (NH_3)_4] [Pt (SCN)_4]$

35. 
How many different species are formed?

36. $\underset{(X)}{C_6H_4} \xrightarrow[(ii)\,Zn/H_2O]{(i)\,O_3} C_3H_2O_3$
no of pi-bond present in X is

37. Number of possible spectral lines which may be emitted in bracket series in H atom, if electrons present in 9th excited level returns to ground level, are

38. 15 g of a solute in 100 g water makes a solution freeze at –1°C. 30 g of a solute in 100 g of water will give a depression in freezing point equal to

39. The enthalpies of neutralization of a weak acid HA & a weak acid HB by NaOH are - 6900 Cal/equivalent & - 2900 Cal/equivalent respectively. When one equivalent of NaOH is added to a solution containing one equivalent of HA & one equivalent of HB, the enthalpy change was - 3900 Calories. If base distributed between HA & HB in X : Y then find value of X+Y.

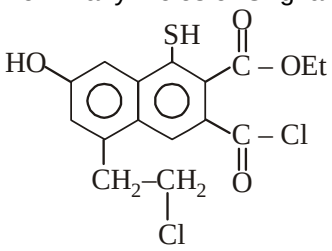
40. 50 mL of 0.1M $CuSO_4$ solution is electrolyzed with a current of 0.965 A for a period of 200 sec. The reactions at electrodes are:
Cathode : $Cu^{2+} + 2e^- \rightarrow Cu(s)$ Anode : $2H_2O \rightarrow O_2 + 4H^+ + 4e$.
Assuming no change in volume during electrolysis, calculate the decimolar concentration of SO_4^{2-} at the end of electrolysis.

41. If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å , 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively. Calculate the average of co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.

Space for Rough work

42. Which of the following statements is (are) true about resonance.
(a) Resonance is an intramolecular process.
(b) Resonance involves delocalization of both s and p electrons.
(c) Resonance involves delocalization of p electrons only.
(d) Resonance decreases potential energy of a molecule.
(e) Resonance has no effect on the potential energy of a molecule.
(f) Resonance is the only way to increase molecular stability.
(g) Resonance is not the only way to increase molecular stability.
(h) Any resonating molecule is always more stable than any nonresonating molecule.
(i) The canonical structure explains all features of a molecule.
(j) The resonance hybrid explains all features of a molecule.
(k) Resonating structures are real and resonance hybrid is imaginary.
(l) Resonance hybrid is real and resonating structures are imaginary.
(m) Resonance hybrid is always more stable than all canonical structures.

43. How many moles of Grignard reagent will be required by one mole of given compound?



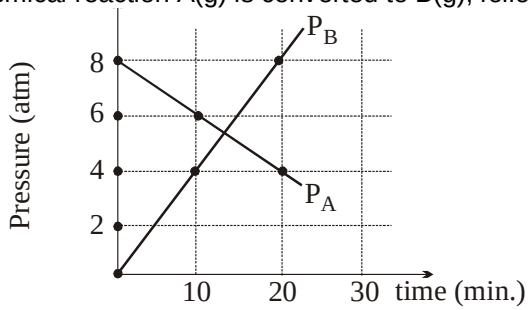
Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

44. KF has NaCl structure. What is the distance between K^+ and F^- in KF in angstrom if density of KF is 2.48 gm/cm^3.

45. In a chemical reaction A(g) is converted to B(g), following observation is made.



Time of completion for the reaction (in min) is

46. A current of 0.5 A is passed through molten $AlCl_3$ for 40 min. Calculate the mass of aluminium in gram deposited at cathode.

Space for Rough work

Mathematics PART – III

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

47. Let f(x) be defined on [0, 1] such that $f(x) = \min \left| x - \frac{1}{2^n} \right|$; n = 0, 1, 2……..then area under f(x) is

(A) $\frac{1}{12}$ (B) $\frac{1}{24}$

(C) $\frac{1}{6}$ (D) $\frac{1}{36}$

48. Consider the region R defined as

$$R = \left\{ (x,y) \left| \begin{matrix} x^2 + y^2 \le & 100 \\ \sin(x+y) \ge & 0 \end{matrix} \right. \right\}$$

What is the area of R;

(A) 25π (B) 40π

(C) 50π (D) 32π

49. The $\lim_{x \to 1} \frac{\left(\log(1+x) - \log 2\right)\left(3.4^{x-1} - 3x\right)}{\left(\left(7+x\right)^{\frac{1}{3}} - \left(1+3x\right)^{\frac{1}{2}}\right)\sin \pi x}$ is

(A) $\frac{2}{3\pi}\left(\log 4 - \frac{1}{2}\right)$ (B) $\frac{4}{3\pi}\left(\log 4 - \frac{2}{3}\right)$

(C) $\frac{9}{4\pi}\left(\log 4 - 1\right)$ (D) $\frac{4}{3\pi}\left(\log 2 - \frac{2}{3}\right)$

Space for Rough work

(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. Let $\vec{a} = 2\hat{i} + \hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 4\hat{i} - 3\hat{j} + 7\hat{k}$. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ be a vector such that $\vec{r} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{r}.\vec{a} = 0$. The magnitude of x + y + z is
(A) 3 (B) 5
(C) 7 (D) 9

51. If ω is one of the angle between normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point whose eccentric angle θ & $\frac{\pi}{2} + \theta$. then $\frac{2\cot\omega}{\sin 2\theta}$ is
(A) $-\frac{e^2}{\sqrt{1-e^2}}$ (B) $\frac{e^2}{\sqrt{1+e^2}}$
(C) $\frac{e^2}{\sqrt{1-e^2}}$ (D) $\frac{e^2}{1+e^2}$

52. If the point p on the parabola $y^2 = 4ax$ for which $|PR - PQ|$ is maximum. Where $R \equiv (-a, 0), Q(0, a)$ is represented as (k_1a, k_2a) then find the value of $(k_1 + k_2)$.
(A) 3 (B) 2
(C) 4 (D) 6

53. Let two tangents $3x - 4y + 20 = 0$ & $x - y - 3 = 0$ of a parabola intersect the tangents at point P(0, 5) and Q(3, 0) respectively, then
(A) length of latus rectum is $24 / \sqrt{34}$
(B) Equation of axis is $3x - 5y - 33 = 0$
(C) focus is (6, -3)
(D) Equation of tangent at vertex 5x + 3y = 15

54. If $f(x) = e^{[\cot x]}$, where [y] denotes the greatest integer less than or equal to y, then
(A) $\lim_{x \to \frac{\pi}{2}^+} f(x) = 1$ (B) $\lim_{x \to \frac{\pi}{2}^+} f(x) = \frac{1}{e}$
(C) $\lim_{x \to \frac{\pi}{2}^-} f(x) = \frac{1}{e}$ (D) $\lim_{x \to \frac{\pi}{2}^-} f(x) = 1$

Space for Rough work

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 55 & 56

In a ΔABC, suppose that $\tan\frac{A}{2}$, $\tan\frac{B}{2}$, $\tan\frac{C}{2}$ are in harmonic progression.

55. What is the minimum possible value of $\cot\frac{B}{2}$?

(A) 1 (B) $\sqrt{2}$

(C) $\sqrt{3}$ (D) 2

56. What is the maximum possible value of angle B?

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

57. The no of values of z satisfying $|z-2i|=2$ and $|z-1-3i|+|z+1-3i|=2\sqrt{2}$ is

58. The minimum value of $|f(x)|$ where

$$f(x)=\frac{\left(x+\frac{1}{x}\right)^6-\left(x^6+\frac{1}{x^6}\right)-2}{\left(x+\frac{1}{x}\right)^3+\left(x^3+\frac{1}{x^3}\right)}$$

Space for Rough work

59. If A is 3×4 matrix and B is 4×3 matrix and det(AB) = 3, then find det(BA)

60. A variable circle S of radius r cuts a rectangular hyperbola H in four distinct point A, B, C and D. If O is the centre of H then the value of
$\lambda = \dfrac{OA^2 + OB^2 + OC^2 + OD^2}{r^2}$ is

61. Let S be the sum of all x in the internal $[0, 2\pi]$ such that $3\cot^2 x + 8\cot x + 3 = 0$. Then the value of $\dfrac{S}{\pi}$ is

62. Let A be $n \times n$ matrix with $|A| = 4$. B is the adjoint of the matrix 2A such that $|B| = 1024$. What is the value of n.

63. The graph xy = 1 is reflected in y = 2x to give the graph $12x^2 + r.xy + sy^2 + t = 0$ $(r + s + t)$ is

64. If the straight line joining origin to the point of intersection of straight lines 4x + 3y = 24 and the curve $(x- 3)^2 + (y - 4)^2 = c^2$ are at right angle then find the value of $|c|$.

65. The value of $\int_0^2 \left[x + \left[x + [x]\right]\right] dx$ where [.] is greatest integer function.

66. If f is real valued differentiable function satisfying $|f(x) - f(y)| \le (x - y)^2$, x, $y \in R$ and f(0) = 0 then f(1) is equal to

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

67. If $\sum_{r=1}^{n} T_r = \dfrac{n(n+1)(n+2)}{3}$ then $\lim_{n\to\infty} \sum_{r=1}^{n} \dfrac{2008}{T_r}$ is

68. Word of length 10 are formed using the letters A, B, C, D, E, F, G, H, I, J. Let X be the number of such word where no letter is repeated and y see the number of such word where exactly are letters is repeated twice and no other letter is repeated then y/9x =

69. Find the minimum value of the sum of real no. $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} $(a > 0)$.

Space for Rough work